

Understanding How Email Works

Complete Tutorial: From Alice Sending an Email to Bob Receiving It

Email is one of the oldest and most widely used internet applications.

Every day, billions of emails are exchanged worldwide.

To users, sending an email looks very simple:

1. Open email app
2. Write message
3. Click **Send**

But internally, multiple protocols, servers, and network operations work together to deliver that email.

This tutorial explains the complete email journey from **Alice sending an email to Bob**, including:

- Email architecture
- User agents
- Mail servers
- Protocols
- Ports
- DNS lookups
- SMTP
- POP3
- IMAP
- Security mechanisms
- Full communication sequence

1. What is Email?

Email (Electronic Mail) is a system for sending digital messages over computer networks.

An email system consists of:

- Sender
- Recipient
- Email client
- Mail servers
- Network protocols

Example Scenario

Alice sends an email to Bob.

Alice's email:

alice@gmail.com

Bob's email:

bob@outlook.com

Message:

Hello Bob

Now let us understand how this message travels.

2. Main Components of Email System

Email communication involves several components.

A. User Agent (Mail User Agent - MUA)

The software used by users to send/receive email.

Examples:

- Gmail Web Interface
- Outlook

- Thunderbird
- Apple Mail

Functions of User Agent

It allows users to:

- Compose email
- Send email
- Receive email
- Read email
- Delete email
- Organize mailbox

Alice Uses

Gmail Web App

Bob Uses

Outlook Client

B. Mail Server

A computer that processes and stores emails.

Mail servers perform:

- Receiving emails
- Forwarding emails
- Delivering emails
- Storing emails

Types of Mail Servers

SMTP Server

Handles sending

Incoming Mail Server

Handles receiving

Uses:

- POP3
- IMAP

C. DNS Server

Used to locate recipient mail server.

Email relies heavily on DNS.

3. Email Protocols

Email uses multiple protocols.

A. SMTP

Simple Mail Transfer Protocol

Used for:

Sending emails

SMTP Port Numbers

Type	Port
SMTP	25
SMTP Submission	587
SMTPS	465

B. POP3

Post Office Protocol v3

Used for:

Downloading emails from server to client

POP3 Ports

Type	Port
POP3	110
POP3 Secure	995

C. IMAP

Internet Message Access Protocol

Used for:

Accessing email while keeping it on server

IMAP Ports

Type	Port
IMAP	143
IMAPS	993

D. DNS (MX Records)

Used to locate recipient mail server

Port:

53

4. Important Email Concepts

Mailbox

Storage area for incoming mail.

Message Queue

Temporary holding area for outgoing mail.

MX Record

DNS record that specifies mail server for domain.

Example:

outlook.com → mail.protection.outlook.com

Envelope vs Message

Envelope

Routing information

Like postal address

Message

Actual content

Subject + Body + Attachments

5. Full Email Journey: Alice Sends Email to Bob

Let us trace every step.

Step 1: Alice Composes Email

Alice opens Gmail.

Writes:

To:

bob@outlook.com

Subject:

Meeting

Body:

Hi Bob, let's meet tomorrow.

Clicks:

Send

Step 2: User Agent Sends Email to Alice's SMTP Server

Alice's email client connects to Gmail SMTP server.

Example server:

smtp.gmail.com

Uses:

SMTP

Port:

587 (usually)

Connection Sequence

TCP Three-Way Handshake

SYN

SYN-ACK

ACK

Connection established.

TLS Handshake

Encryption begins.

Step 3: SMTP Session Starts

Client says:

EHLO

Server responds:

250 OK

Authentication happens.

Alice logs in.

Client sends:

AUTH LOGIN

Credentials transmitted securely.

Step 4: Email Submitted to Gmail Mail Server

SMTP commands:

MAIL FROM

MAIL FROM:<alice@gmail.com>

RCPT TO

RCPT TO:<bob@outlook.com>

DATA

Client sends email content.

Example:

Subject: Meeting

Hi Bob, let's meet tomorrow.

Server responds:

250 Message accepted

Email enters Gmail outgoing queue.

Step 5: Gmail Needs Bob's Mail Server Address

Gmail extracts Bob's domain:

outlook.com

Now Gmail must locate Outlook's mail server.

It performs DNS lookup.

Step 6: DNS MX Record Lookup

Gmail queries DNS:

MX outlook.com

DNS responds:

mail.protection.outlook.com

This is Bob's receiving mail server.

Why MX Record?

Websites use:

A record

Email uses:

MX record

Because email delivery requires dedicated mail servers.

Step 7: Gmail Connects to Outlook Mail Server

Gmail SMTP server opens connection to Outlook SMTP server.

Uses:

SMTP

Usually Port:

25

TCP handshake occurs.

SMTP session begins.

Step 8: Server-to-Server SMTP Transfer

Gmail SMTP sends:

```
MAIL FROM:<alice@gmail.com>
```

```
RCPT TO:<bob@outlook.com>
```

```
DATA
```

Outlook server accepts.

Response:

```
250 OK
```

Email transferred.

Step 9: Bob's Mail Server Stores Email

Outlook mail server stores message in Bob's mailbox.

Email waits there until Bob checks mail.

Step 10: Bob Opens Email Client

Bob opens Outlook app.

Client connects to incoming mail server.

Using:

IMAP or POP3

If Using IMAP

Port:

993

Secure connection

If Using POP3

Port:

995

Secure connection

Step 11: Bob Authenticates

Bob logs in.

Server verifies credentials.

Step 12: Bob Downloads or Views Email

If IMAP:

Email remains on server

If POP3:

Email usually downloads locally

Bob reads:

```
Hi Bob, let's meet tomorrow.
```

Delivery complete.

Complete Sequence Diagram

Alice writes email
↓
Alice's User Agent
↓
SMTP to Gmail Server
↓
DNS MX Lookup
↓
Locate Outlook Mail Server
↓
SMTP Transfer
↓
Bob's Mail Server
↓
Stored in Mailbox
↓
Bob's Client connects
↓
IMAP/POP3 Retrieval
↓
Bob reads email

6. Detailed Protocol Flow

Sending Side

Alice Client → Gmail SMTP

Protocol:

SMTP

Port:

587

Server-to-Server

Gmail → Outlook

Protocol:

SMTP

Port:

25

Receiving Side

Bob Client → Outlook

Protocol:

IMAP / POP3

Ports:

993 / 995

7. SMTP Commands Explained

HELO / EHLO

Introduce client

MAIL FROM

Sender address

RCPT TO

Recipient address

DATA

Actual message

QUIT

Close session

Example:

```
EHLO
MAIL FROM
RCPT TO
DATA
QUIT
```

8. IMAP vs POP3

Feature	POP3	IMAP
Downloads mail	Yes	Optional
Keeps on server	Usually no	Yes
Sync multiple devices	No	Yes
Modern usage	Less common	Very common

Today most systems use:

IMAP

9. Email Security Mechanisms

Modern email uses several protections.

A. TLS Encryption

Protects email in transit

B. SPF

Verifies sending server

C. DKIM

Digitally signs email

D. DMARC

Defines handling of suspicious mail
These reduce spoofing and phishing.

10. What Happens if Server is Offline?

Mail server queues email.

Retries later.

Typical retry schedule:

- 15 min
- 30 min
- 1 hour
- Several hours

Eventually may bounce.

11. What is SNMP's Role?

You asked about SNMP.

Important clarification:

SNMP is **not used for sending email**.

SNMP stands for:

Simple Network Management Protocol

It is used for:

- Monitoring mail servers
- Checking server health
- CPU usage
- Disk space
- Network status

Example:

Admin checks:

- Is SMTP server running?
- Queue size?
- Packet loss?

SNMP Port:

Protocol	Port
----------	------

SNMP	161
------	-----

SNMP Trap	162
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So:

SNMP supports management, not email delivery.

12. Real-Life Postal Analogy

Alice writes letter

↓

Post office accepts letter

↓

Finds Bob's city

↓

Transfers to Bob's local post office

↓

Stores in Bob's mailbox

↓

Bob checks mailbox

That is exactly how email works.

13. Important Ports Summary

Protocol	Port	Purpose
DNS	53	Mail server lookup
SMTP	25	Server transfer
SMTP Submission	587	Client sending
SMTPS	465	Secure sending
POP3	110	Receive
POP3S	995	Secure receive
IMAP	143	Access mail
IMAPS	993	Secure access
SNMP	161	Monitoring

Final Summary

When Alice sends email to Bob:

1. Alice writes email
2. User agent connects to SMTP server
3. SMTP session starts
4. Email submitted
5. Sender server performs DNS MX lookup
6. Recipient mail server located
7. Server-to-server SMTP transfer
8. Email stored in Bob's mailbox
9. Bob connects using IMAP/POP3
10. Bob reads email

All of this usually happens within seconds.